Evolution of Computing

Sequential Computing (Classical): A single CPU executes instructions one by one, in order. Performance was traditionally improved by increasing clock speed, but physical constraints (power/heat) led to the adoption of parallel architectures.

Parallel Computing: Multiple processors within a **single computer** work simultaneously on subtasks of a single problem, usually sharing memory.

Distributed Computing: Multiple **independent computers** connected via a network collaborate to solve a shared goal. They do not share memory and communicate via message passing.

Parallel & Distributed Computing Differences

Architecture: Parallel computing connects processors within one system (tightly coupled), whereas distributed computing connects multiple distinct computers over a network (loosely coupled).

Memory Usage: Parallel systems usually rely on shared memory, allowing processors to access the same data instantly. Distributed systems use distributed memory, meaning each computer has its own, requiring data to be passed between nodes.

Communication & Speed: Parallel systems use fast, internal buses for communication. Distributed systems rely on slower network communication, often leading to higher latency.

Scalability: Distributed systems are highly scalable (easily adding more computers), whereas parallel systems are limited by the physical hardware limits of a single machine.

Fault Tolerance: Distributed computing is generally more robust; if one node fails, the rest continue to work. Parallel systems are susceptible to failures if the main system fails.

Use Cases: Parallel computing is used for tasks requiring immense speed, such as, for example, climate modeling or, scientific simulations. Distributed computing is best for applications requiring high availability or wide geographic distribution, like web services and distributed databases.

Parallel & Distributed Computing Differences

Feature	Parallel Computing	Distributed Computing
Coupling	Tightly coupled	Loosely coupled
Memory	Shared Memory	Distributed Memory
Nodes	One computer (multi-core)	Multiple independent computers
Network	High-speed bus/backplane	Network (LAN/WAN/Internet)
Speed	Extremely fast (low latency)	Dependent on network (higher latency)
Primary Goal	Speed up computing	Scalability and reliability

Advantages of Parallel Computing

- It saves time and money as many resources working together will reduce the time and cut potential costs.
- It can be impractical to solve larger problems on Serial Computing.
- It can take advantage of non-local resources when the local resources are finite.
- Serial Computing ‘wastes’ the potential computing power, thus Parallel Computing makes better work of hardware.

Types of Parallelism

Bit-level Parallelism

It is the form of parallel computing which is based on the increasing processor's size. It reduces the number of instructions that the system must execute in order to perform a task on large-sized data.

Example: Consider a scenario where an 8-bit processor must compute the sum of two 16-bit integers. It must first sum up the 8 lower-order bits, then add the 8 higher-order bits, thus requiring two instructions to perform the operation. A 16-bit processor can perform the operation with just one instruction.

Types of Parallelism

Instruction-level parallelism

A processor can only address less than one instruction for each clock cycle phase. These instructions can be re-ordered and grouped which are later on executed concurrently without affecting the result of the program. This is called instruction-level parallelism.

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Task Parallelism

Task parallelism employs the decomposition of a task into subtasks and then allocating each of the subtasks for execution. The processors perform execution of sub tasks concurrently.

Why parallel computing?

- The whole real world runs in dynamic nature i.e. many things happen at a certain time but at different places concurrently. This data is extensively huge to manage.
- Real world data needs more dynamic simulation and modeling, and for achieving the same, parallel computing is the key.
- Parallel computing provides concurrency and saves time and money.
- Complex, large datasets, and their management can be organized only and only using parallel computing's approach.
- Ensures the effective utilization of the resources. The hardware is guaranteed to be used effectively whereas in serial computation only some part of hardware was used and the rest rendered idle.
- Also, it is impractical to implement real-time systems using serial computing.

Applications of Parallel Computing

- Data bases and Data mining.
- Real time simulation of systems.
- Science and Engineering.
- Advanced graphics, augmented reality and virtual reality.

Limitations of Parallel Computing

- It addresses such as communication and synchronization between multiple sub-tasks and processes which is difficult to achieve.
- The algorithms must be managed in such a way that they can be handled in the parallel mechanism.
- The algorithms or program must have low coupling and high consistency. But it's difficult to create such programs.
- More technically skilled and expert programmers can code a parallelism based program well.

Future of Parallel Computing

- The computational graph has undergone a great transition from serial computing to parallel computing.
- Tech giant such as Intel has already taken a step towards parallel computing by employing multicore processors.
- Parallel computation will revolutionize the way computers work in the future, for the better good.
- With all the world connecting to each other even more than before, Parallel Computing does a better role in helping us stay that way.
- With faster networks, distributed systems, and multi-processor computers, it becomes even more necessary.